

Two routes to a simple input

Can one or two sine waves do what the earthquake record does to the wall?

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The question

One test wall, one record, two ways to replace the record

The goal

Replace the 17 s earthquake record by one or two sine waves that give the model the same wall movement.

Cheaper, and the wave can be tuned to the state of the wall.

The test case

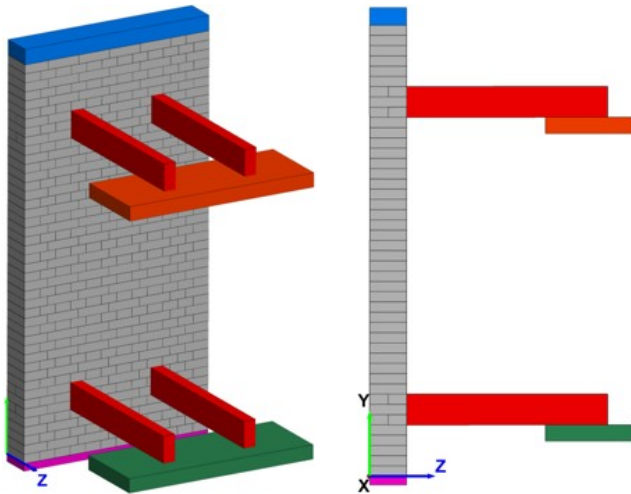
SAFEGo wall US-1, run 24 ($FR76 \times 1.75$).
Measured: 29.5 mm out-of-plane at 2.06 m, 29.5 kN base shear.

Model: 3DEC, rigid blocks, all damage in the joints.

The two routes

Route 1 (Eleni): one sine wave with the record's peak velocity.

Route 2 (Ihsan): two sine waves, one at the first period of the wall, one at the softened period.



How we judge a route

- First against the record on the same model: does the wave move the model as much as the record does?
- Then against the test: 29.5 mm and 29.5 kN at run 24.
- Two checks, in that order. A route can pass the first and still not the second.

Before the routes: pushing the model over, two ways

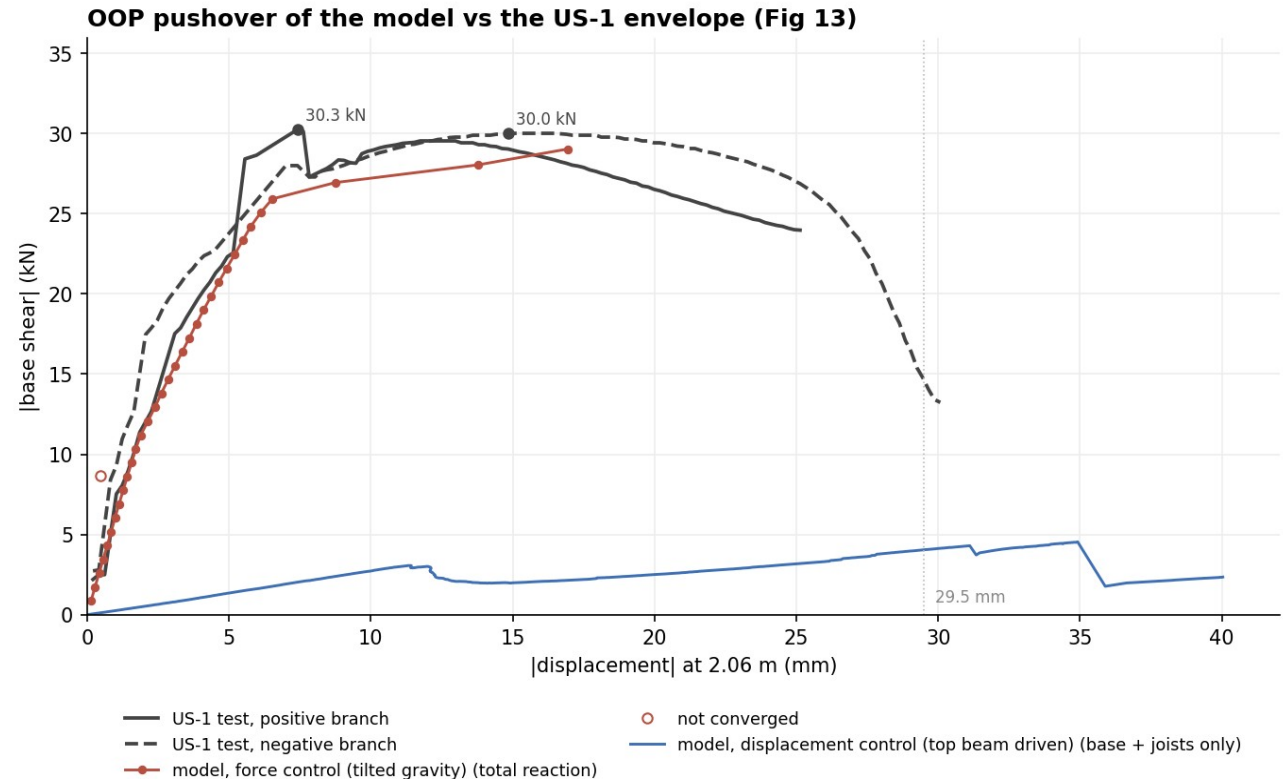
The static force-displacement curve of the model, needed for Route 2 and for the check against the test

Force control (tilted gravity)

Load grows step by step until the model cannot hold it. Follows the test curve up to the peak: 29.0 kN at 17 mm; test 29.5 kN. Rocking starts at 7–8 mm. Stops at the peak: no equilibrium on the falling branch.

Displacement control (pushing the top beam)

Top beam moved at a fixed speed, base shear read. Only 3–5 kN at 30 mm. Pushing the restrained top beam measures sideways sliding, not rocking. Not a capacity curve.



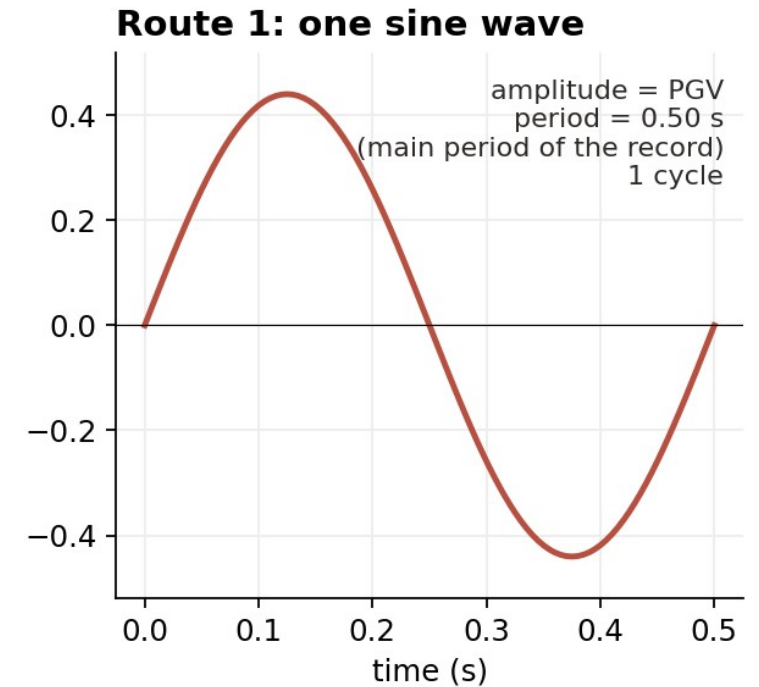
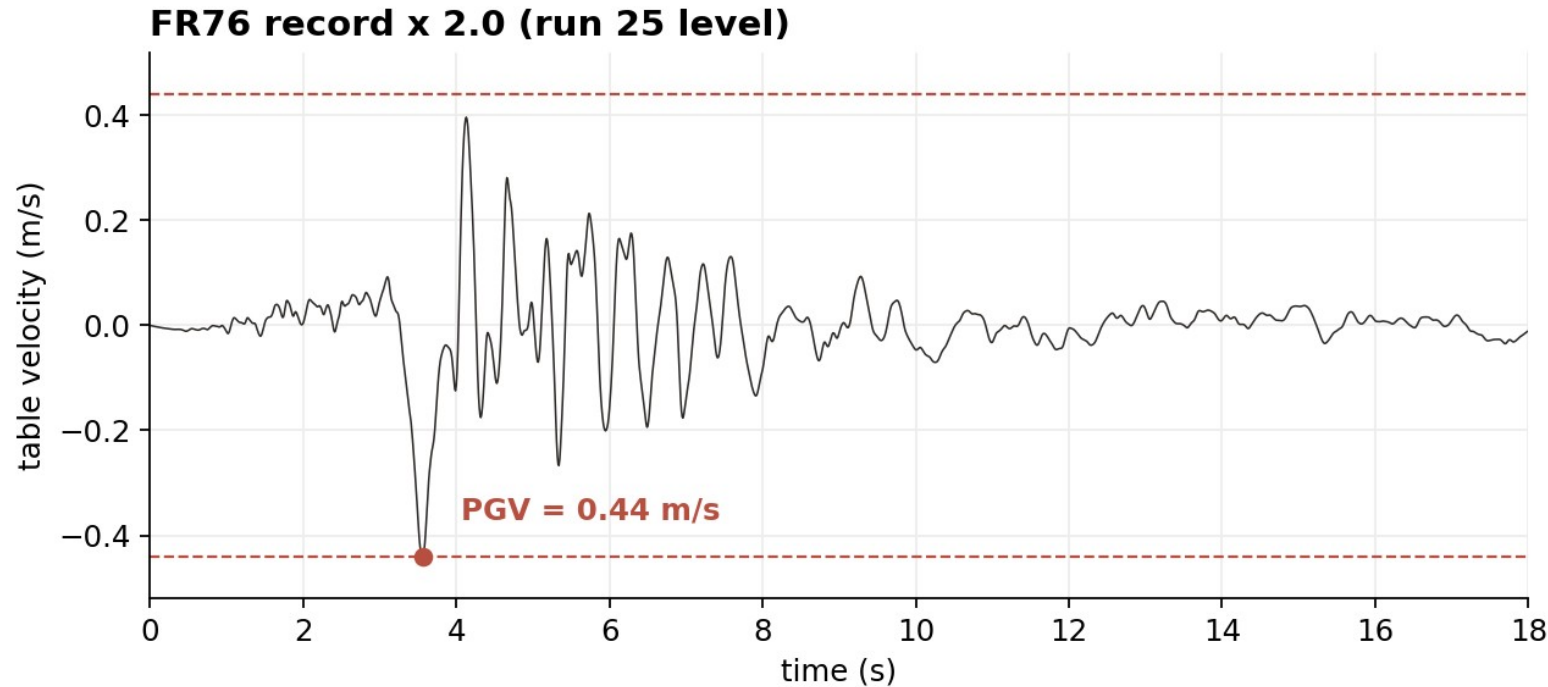
Model curves against the test envelope (Fig 13). Force control: total reaction; displacement control: base + joists only. Hollow point = step that did not converge.

Force control: right peak, no falling branch. Top-beam displacement control: wrong curve. New driver (ready): hold the wall at a target displacement with the same tilted-gravity load.

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Route 1: one sine wave with the record's peak velocity

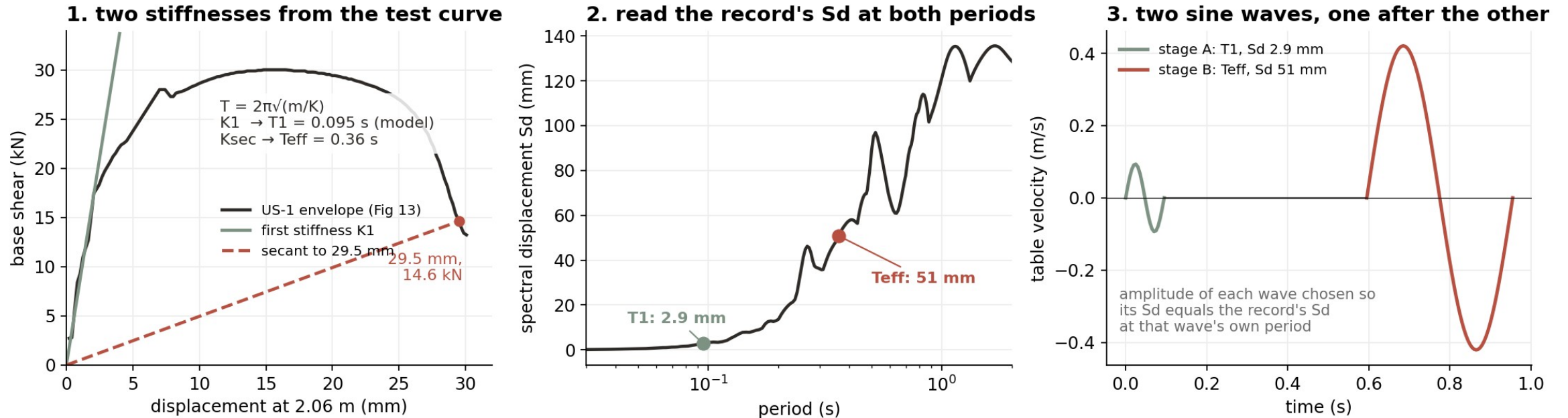
Eleni's route — nothing about the wall goes in



- Amplitude: the peak velocity of the record, times the run's scale factor (0.44 m/s at run-25 level).
- Period: the main period of the record (0.50 s). One cycle.
- Simple and fixed. If it works, no wall information is needed.

4 Route 2: two sine waves, one after the other

Ihsan's route — the wave follows the state of the wall



- Stage A: period = first period of the model (0.095 s). Amplitude set so its Sd equals the record's Sd there (2.9 mm at run-25 level).
- Stage B: period from the secant to 29.5 mm on the test curve, using the envelope force (14.6 kN) and $m = 1635 \text{ kg}$: $T = 0.36 \text{ s}$. Amplitude set so its Sd equals the record's Sd there (51 mm).
- Both stages one cycle, 0.5 s of quiet between them.

Three tests

Same model, same input level, three starting points

Test 1: undamaged wall

Start from the new wall.
Apply Route 1 or Route 2 at run-25 level
(FR76 $\times$ 2.0).

One shot. The wall has no damage when
the wave arrives.

Reference: the record

Drive the model with the real table motion,
all 25 runs in the test order (Strategy F).

This is what a route has to reproduce: the
record's effect on this model.

Test 2: damaged wall

Restart from the model state at the end of
run 23 or run 24 of the record sequence.
Apply Route 2 at the next run's level.

Compare with what the record did in that
next run.

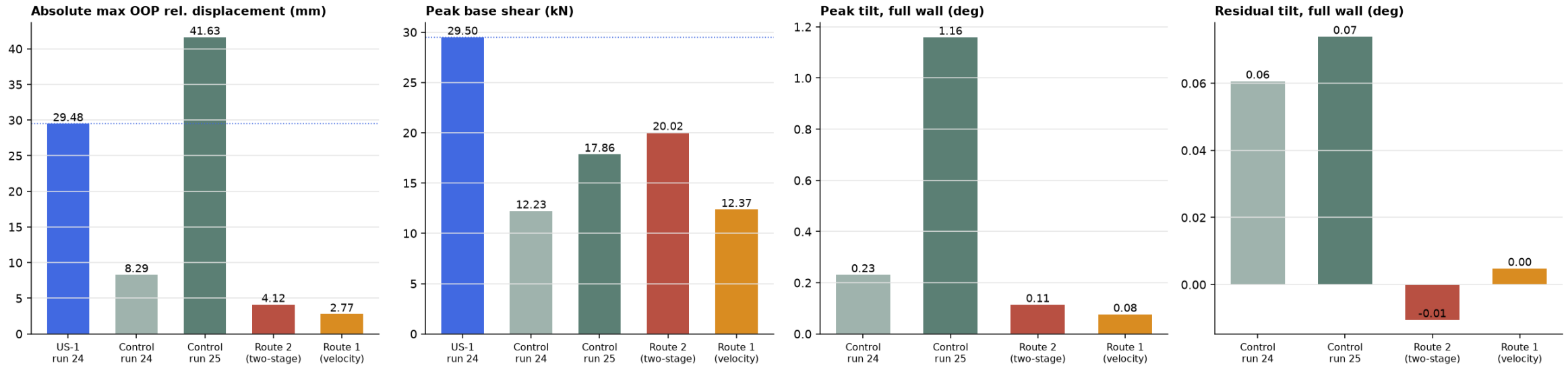
Test 1 asks: can the waves crack the wall? Test 2 asks: once the wall is cracked, do the waves move it as much as the record does?

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Test 1 result: on the undamaged wall both routes do very little

Peak values of the one-shot runs, next to the record on the same model and the test

One-shot routes (FR76 x2.0) vs Strategy F control (runs 24/25) vs US-1 run 24 (base shear = cstav (base+joist only))



Numbers

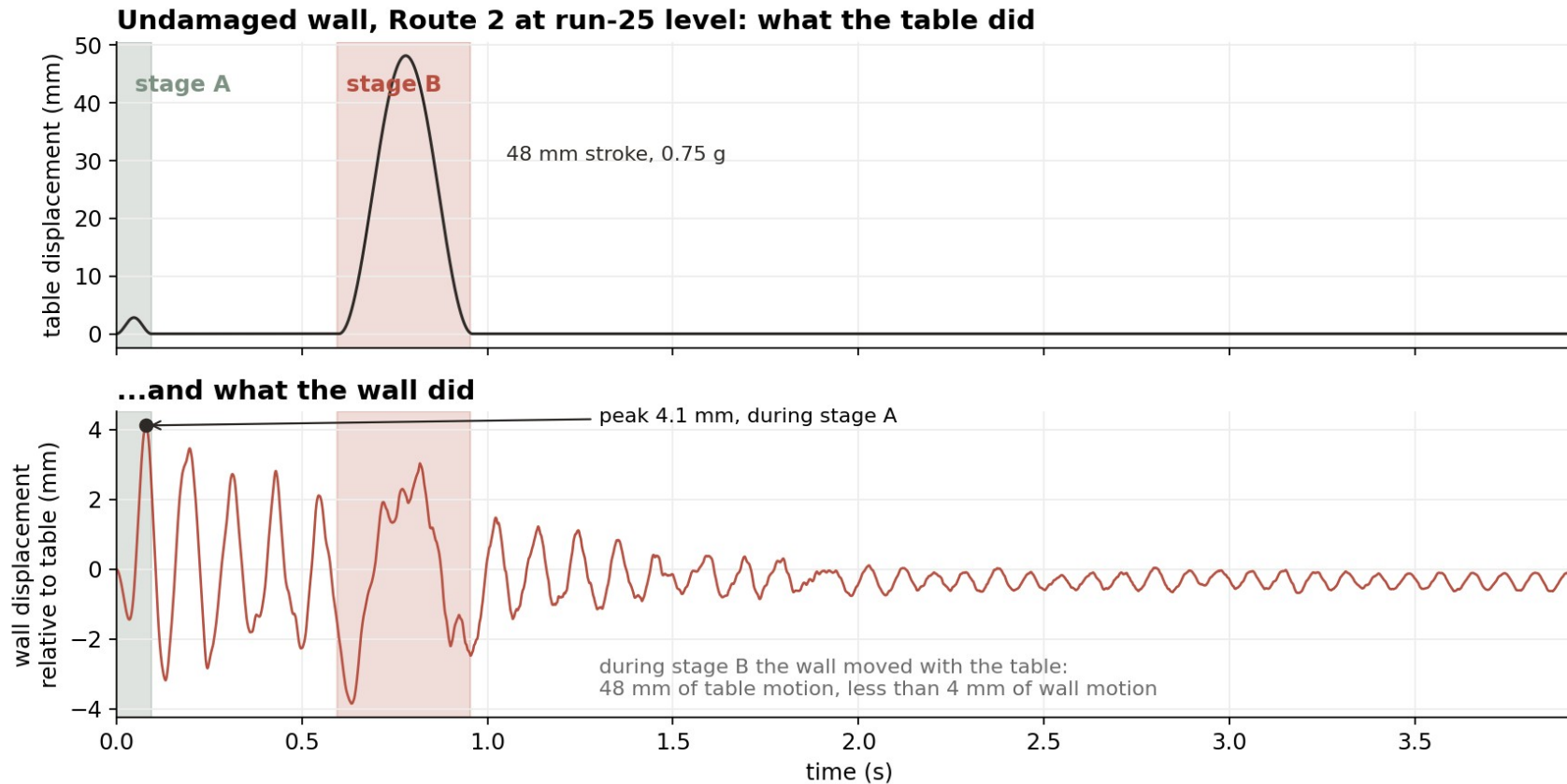
Route 1: 2.8 mm. Route 2: 4.1 mm.
 Record on the model: 8.3 mm at $\times 1.75$, 41.6 mm at $\times 2.0$.
 Test US-1 at run 24: 29.5 mm.

Reading

Neither wave cracks the new wall. The record at $\times 2.0$ does, and then the model moves a lot.
 Base shear is base + joists only in the model (no top-joint share yet), so the force bars are not like-for-like with the test.

Why: stage B needs a wall that is already soft

Route 2 on the undamaged wall, run-25 level



What happened

Stage A (2.9 mm demand at 0.095 s): the wall answers with 4.1 mm. That is the peak.

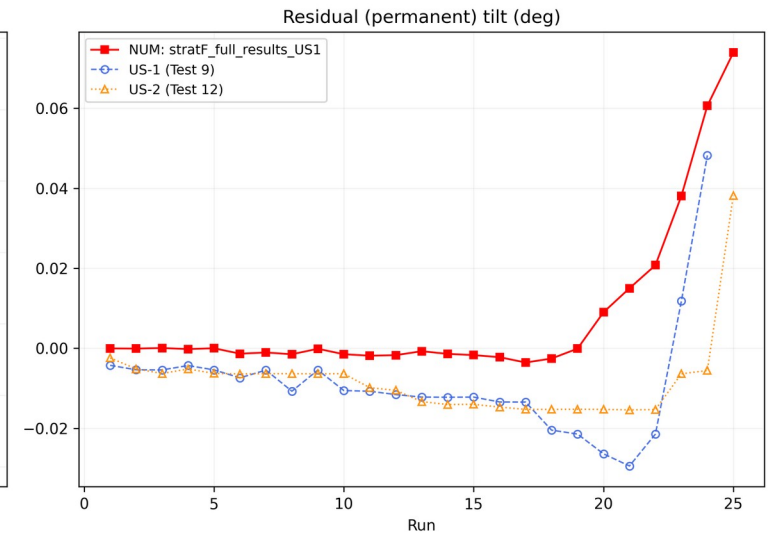
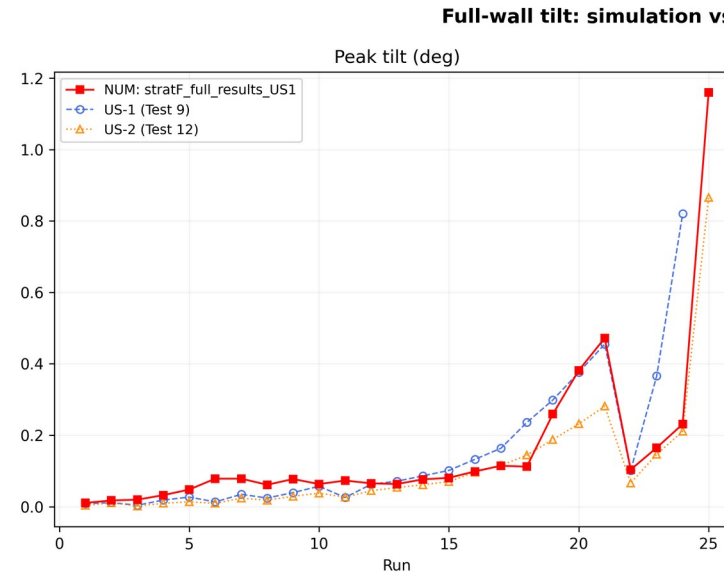
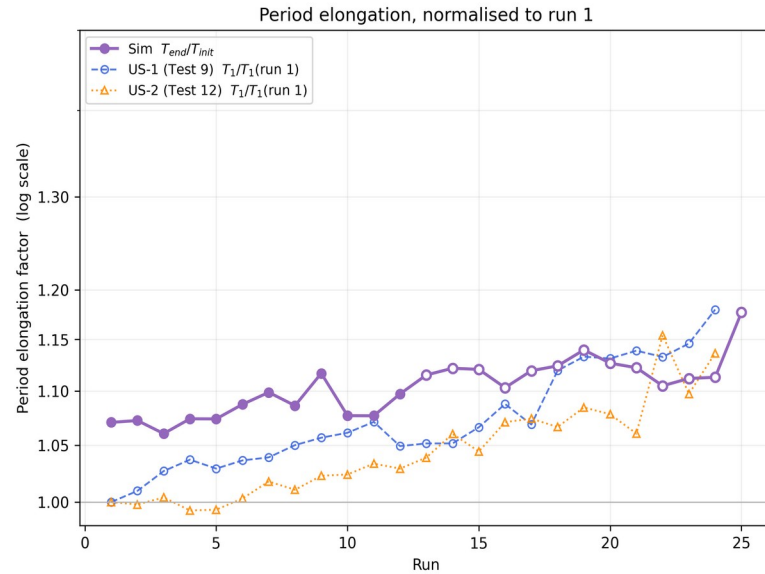
Stage B is a 48 mm, 0.75 g push at 0.36 s. The wall was still at ~0.1 s, so it just rode along with the table.

The 51 mm of S_d at 0.36 s only exists for a wall that is already at 0.36 s.

Stage B is built for a soft wall. On a stiff wall it does nothing. So the next test starts from a wall the record has already cracked.

Reference: period and tilt tell the same story

Model vs both test walls, run by run



Period

The model's period grows about 4% from run 1 to run 24 and 10% by run 25. The test wall grew 17.5% by run 24.

The model softens one step later than the test.

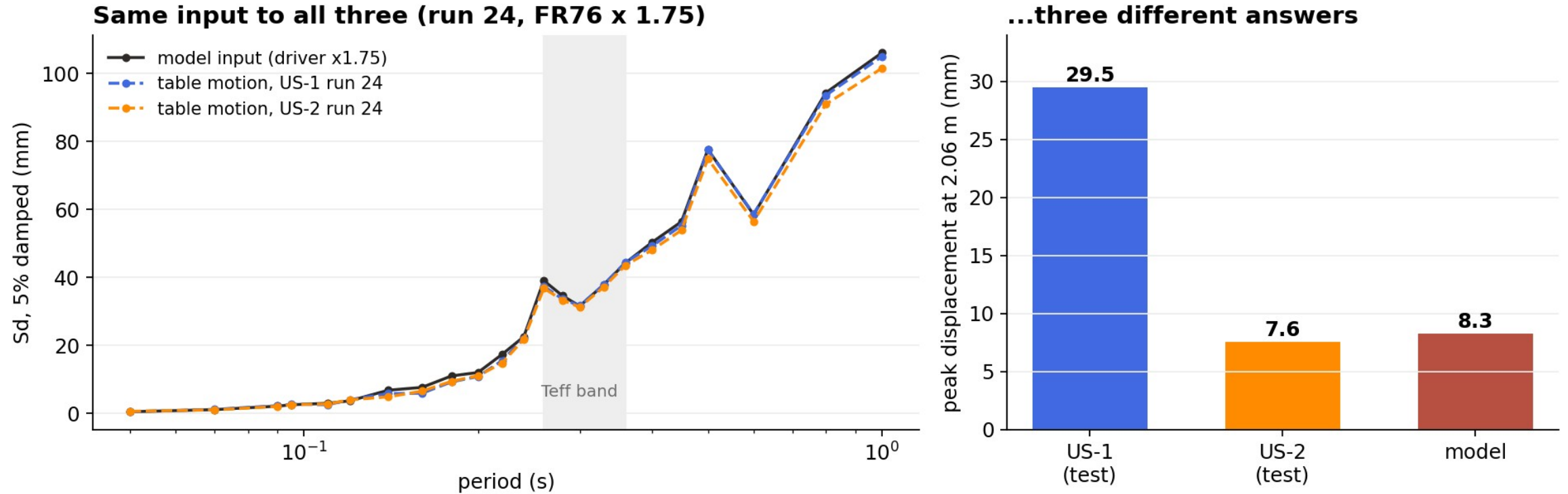
Tilt

Peak tilt follows the test up to run 21 and then lags, like the displacement.

The permanent tilt of the model leans the other way than US-1 over runs 15–22 — to be checked (sign convention or joist side).

9 Same input, two test walls, two answers

Run 24 on US-1 and on US-2 (its twin), and on the model



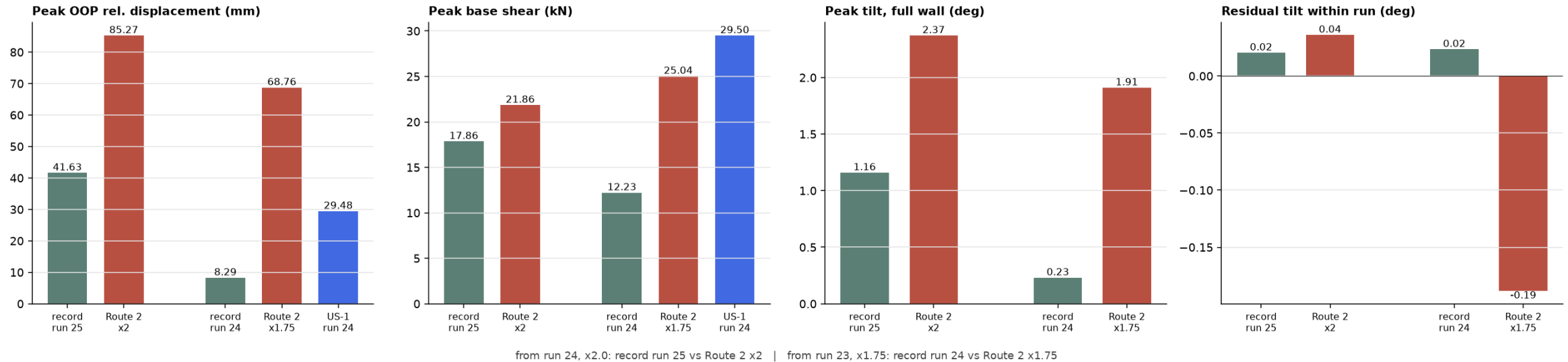
Reading

The table gave both walls the same motion (spectra within 2-5%), and the model got the same input. US-1 moved 29.5 mm, US-2 moved 7.6 mm, the model 8.3 mm. The 29.5 mm target belongs to one of two twin walls; the paper itself reports a large unexplained difference between them.

Test 2 result: on the damaged wall Route 2 does far too much

Restart from the record state, apply Route 2, compare with the record on the same state

Route 2 on restored Strategy F states vs the record on the same states (within-run metrics; shear = cstav (base+joist only))



Numbers

From run 24 at $\times 2.0$: Route 2 gives 85 mm, the record gives 42 mm (2 $\times$).

From run 23 at $\times 1.75$: Route 2 gives 69 mm, the record gives 8 mm (8 $\times$).

Reading

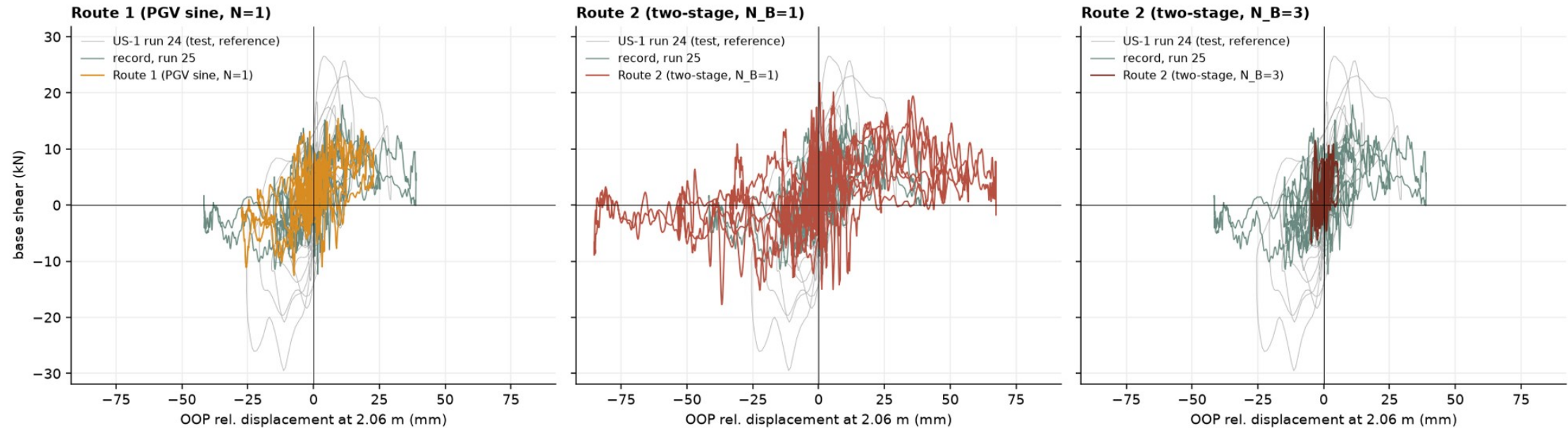
One big sine cycle at 0.36 s is a 48 mm push in 0.18 s. A cracked, rocking wall follows that push almost fully.

The record has the same S_d at 0.36 s, but its pushes are shorter and go both ways, so the wall ends up at half of that.

Same damaged wall, same level: the record vs three waves

Base shear vs displacement, last run (FR76 x2.0) from the end-of-run-24 state

Last run (FR76 x2.0, from the end-of-run-24 state): record vs routes [shear = cstav (base+joists only)]



- The record (run 25) takes the wall to 42 mm.
- Route 1, one wave with the record's peak velocity: about 25 mm. Closest of the three; same loop shape and force level as the record.
- Route 2 with one cycle: 85 mm, twice the record. With three cycles: 8 mm, a fifth of it. Same S_d target, opposite answers.
- All three reach a similar base shear (10–20 kN). They differ in how far the wall rocks, not in how hard it pushes back.

The wave that carries the record's velocity gets closest. The waves that carry its S_d at one period miss in both directions.

Base shear = base + joists only (about 55% of the total). Grey loop: US-1 run 24, reference only — run 25 was not applied to US-1.

What we learned

The two routes bracket the record, from both sides

Undamaged wall

Both routes: 3–4 mm.

Record: 8 to 42 mm.

Too little. The waves do not crack the wall.

Damaged wall

Record: 42 mm.

Route 1: 25 mm. Route 2: 85 mm (1 cycle)
or 8 mm (3 cycles).

The velocity wave comes closest; the Sd
waves miss both ways.

The model itself

Follows the test to 17 mm, sits on the twin
wall US-2 at run 24, cracks one scale step
after US-1.

Input to the model checked against the real
table motion: fine.

Matching Sd at one period fixes the size of the wave but not how hard a single cycle pushes. A rocking wall answers the push, not the Sd.

Next steps

Cheap tests first, then the decisions

1. Spread stage B over 3 cycles

Same S_d target at 0.36 s, three smaller cycles instead of one big one (0.27 g instead of 0.75 g). Run from the same two saved states. Script ready.

2. Route 1 on the damaged wall

Same restart test for the one-wave route, from run 23 and run 24. Script ready.

3. Full model force-displacement curve

Pushover with the falling branch (new driver). Needed for the secant period from the model instead of the test curve, and to compare with the test envelope.

4. Decisions for this meeting

Which wall is the target: US-1, US-2, or the range between them?
Confirmed so far: envelope force for the secant, model first period for stage A.